

Lesson 4: Lists Make

45 minutes

Overview

Using Programming Patterns and a step-by-step approach students make their own version of a Reminder app. At the beginning of the lesson students are able to explore a working version of the app. They are then given the design elements of the app but begin with a blank screen. Students use an Activity Guide to go through the high level steps they should use to develop their app but leaves it to them to decide how to write the code. At the end students submit their apps and answer a free-response question in the style of a Create performance task written response prompt. Student apps and written responses can be assessed using a provided rubric.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- AP - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Build the Reminder App

Wrap Up (5 minutes)

Assessment: Make Project

Objectives

Students will be able to:

- Implement programming patterns with lists to develop a functioning app
- Recognize the need for programming patterns with lists as part of developing a functioning app
- Use debugging skills as part of developing an app
- Write comments to clearly explain both the purpose and function of different segments of code within an app

Preparation

- Review the Free-Response Supports and consider which, if any, supports are suitable for your classroom.
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 6 - Lesson 4 - Lists Make** - Slides [▼ Make a Copy](#)
- **Writing Support - Lists Make** - Handout [▼ Make a Copy](#)

For the students

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** *Imagine you want to build an app for reminders. What information do you think would be stored in a list?*

Discussion Goal: Students may struggle with this question. It's ok if they don't get it right away. The goal is to prime their thinking before they look at the sample app.

Sample Answer: The reminders could be stored in a list.

Remarks

For the past few days, we've learned a lot about using lists to help apps keep track of information. In today's Make Project you'll be practicing using lists and programming patterns with lists to create a functioning Reminders app.

Activity (35 minutes)

Build the Reminder App

Teaching Tip

Supporting Students: While students are working on their apps, circulate the room and check in with students who need a little help. Encourage students to collaborate and discuss bugs with each other.

Debugging: Review with students steps they can use to debug if they get stuck:

- Run the code on turtle mode
- Add the variables to the watcher
- Explain the code to a friend

Group: Make a determination as to whether this project will be completed in pairs or individually. You may even choose to let students decide.

 **Do This:** Have students explore the working Reminder App in Level 1.

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Try the Reminders App

If students are not working in pairs they should still discuss the discuss with a neighbor.

 **Discuss:**

- *How many lists do you think are needed to make this app work?*
 - Only one list is needed. It stores all of the reminders.
- *What programming patterns with lists do you think you'll need to use?*
 - This app uses the List Scrolling pattern.

Remarks

Now let's build the this app. The screen has been set up for you - it's your job to add the code!

 **Do This:** Direct students to level three where they complete the Reminder App. An optional Activity Guide is provided if students would like guidance in creating the app. The most relevant programming pattern is displayed on a slide. Review this pattern quickly with students, if needed.

 **Submit:** Encourage students to check the rubric on the last page of the Activity Guide before submitting.

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Make the Reminders App

Teaching Tip

Adapting Make Lessons - You may decide that you'd like to have your students write the code for a modified or entirely different app using the same skills. This might help you differentiate your instruction or provide you with a way to reassess students. Check out the [Customize Make Lessons: How-To-Guide](#) for guidance on how to get started with this.

Wrap Up (5 minutes)

Remarks

Great work on your Reminder app today. Before wrapping up, we will practice writing for the style of questions that are asked as part of the Create performance task on the AP exam. Respond to the question in Step 5 in your activity guide, and submit it along with your project.

 **Do This:** Complete the written response in Step 5 of the Activity Guide.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Interface Support: For students who did not fully complete their app, provide them with this snippet from the exemplar code.

Assessment: Make Project

Use the rubric provided with the project to assess student projects.

Assessment Opportunity

Student responses can be used as a formative assessment to evaluate their understanding of connections between user interface features and functionality. Review their explanations for clarity, accuracy, and alignment with the AP exam's expectations. The provided rubric can be used to guide this review, and to help provide students with feedback to guide their learning and improvement.



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